



UNIVERSITI TUNKU ABDUL RAHMAN (UTAR)
FACULTY OF INFORMATION AND COMMUNICATION
TECHNOLOGY (FICT)

Teaching Plan

Part A: Course Information

1.	Course Code & Course Title:	UCCD3243 Server-Side Web Applications Development												
2.	Year of Study:	3												
3.	Credit Hour:	3												
4.	Lecturer:	Nur Athirah Nabila Mohd Idros												
5.	Tutor:	Nur Athirah Nabila Binti Mohd Idros Dr. Altahir Abdalla Altahir Mohammed Ts. Dr. Suthashini a/p Subramaniam												
6.	Year and Trimester:	June 2026 Trimester												
7.	Synopsis:	This unit equips students with the knowledge and skills that are required to develop Web based applications. The topics covered include setting up and configuring Web servers/application servers, technology features for handling client requests and generating server responses. Further topics include technologies for maintaining state in applications, separating content from presentation, and connecting and manipulating databases and securing a web site. Apart from that, topics covering a range of relevant web technologies, tool and methods will also be included.												
8.	Course Learning Outcome (CLO):	<table><tr><td>Course Outcomes: After completing this course, students will be able to:</td><td></td></tr><tr><td>Course Learning Outcomes (CLO):</td><td>Domain & Taxonomy Level¹</td></tr><tr><td>CLO 1 – Assess a range of relevant web technologies, tools and methods and approaches for building Web based applications.</td><td>C5</td></tr><tr><td>CLO 2 – Build web based applications that handle client requests and generate server responses.</td><td>P2</td></tr><tr><td>CLO3 – Appraise web connectivity of web applications that access and manipulate data stored in databases.</td><td>C4</td></tr><tr><td>CLO4 – Analyze methods to maintain a secure web site.</td><td>C4</td></tr></table> ¹Domain and Taxonomy Level – Cognitive (C), Level 1-6; Affective (A), Level 1-5; Psychomotor (P), Level 1-5	Course Outcomes: After completing this course, students will be able to:		Course Learning Outcomes (CLO):	Domain & Taxonomy Level ¹	CLO 1 – Assess a range of relevant web technologies, tools and methods and approaches for building Web based applications.	C5	CLO 2 – Build web based applications that handle client requests and generate server responses.	P2	CLO3 – Appraise web connectivity of web applications that access and manipulate data stored in databases.	C4	CLO4 – Analyze methods to maintain a secure web site.	C4
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9.	Reading List:	Main References: 1. Jon Duckett (2022). PHP & MySQL: Server-side Web Development, Wiley, 1st Edition Additional References: 1. Joel Murach, Ray Harris. (2022). Murach's PHP and MySQL, Mike Murach & Associates, Incorporated, 4th Edition 2. Paul Gibbs (2021). PHP Tutorials: Programming with PHP and MySQL: Learn PHP 7 / 8 with MySQL Databases for Web Programming, 5th edition
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Part B: Constructive Alignment Table

No.	CLO	Programme Learning Outcomes (PLO)	Domain & Taxonomy Level [#] (e.g. A2 / C3 / P5)	Delivery Methods (e.g. Lecture, Tutorial, Practical, Case Study, Site Visit, Role Play, Class Discussion)	Assessment Methods & Mark Breakdown*							
					Midterm Test	Quiz	Assignment	Practical	Final Assessment Q1	Final Assessment Q2	Final Assessment Q3	Final Assessment Q4/Q5
					15%	5%	15%	15%	12.5 %	12.5 %	12.5 %	12.5 %
1	CLO1	3	C5	Lecture, Practical	√	√			√			√
2	CLO2	2	P2	Lecture, Practical			√	√				
3	CLO3	3	C4	Lecture, Practical						√		
4	CLO4	3	C4	Lecture, Practical							√	

[#] Domain – Affective (A), Cognitive (C), Psychomotor(P); Taxonomy Level – A (Level 1 – 5), C (Level 1 – 6), P (Level 1 – 5)

*Assessment methods and the mark breakdown must be tally with the teaching workload approved by Senate. ONE type of Coursework assessment method can be mapped to ONLY one CLO.

Assessment Methods:

Continuous Assessment	CLO	Percentage (%)	F2F	NF2F	Total SLT
Quiz	1	5	1	1	2
Midterm	1	15	1	4	5
Assignment	2	15	1	4	5
Practical	2	15	2	2	4
Final Assessment	CLO	Percentage (%)	F2F	NF2F	Total SLT
Final Examination	1,3,4	50	2	14	16

Part C: Lecture, Tutorial/Practical and Assessment Plan

Week	Lecture Topic	Practical Topic	Assessments / Specific Tasks
1 15/6/26 - 21/6/26	Introduction to Web Client-Server - Internet application: World Wide Web (WWW) - Client-Server Model - Browser Communication with Server - Static and Dynamic Websites - Multi-Tiered Approach in Development of Applications - Client-Side and Server-Side Scripting - PHP Fundamentals	**Practical classes start on Week 2	**17/6 (Wed) – Awal Muharram
2 22/6/26 - 28/6/26	Server-Side Handling - HTTP Concepts - HTTP Status Codes - HTTP Request Structure, Response Structure - HTTP and PHP for Request Handling - PHP Superglobal - PHP Functions - PHP Action and Method Attributes - PHP GET and POST Method	Practical 1: Local Web Development Environment Setup	Release of Group Assignment (4 members in a group)
3 29/6/26 - 5/7/26	Database Technology Driven Web Applications - Database Technology - MySQL - Structured Query Language (SQL) - Database Connection using PHP - Create, Read, Update, Delete (CRUD) Overview - User Interaction and Data Manipulation	Practical 2: PHP Fundamentals and Server-Side Handling	
4 6/7/26 - 12/7/26	Maintaining State in Web Applications - Stateful and Stateless Web Application - Cookie-Based State Management in PHP - Session Handling in PHP - Best Practices for Session Management - User Authentication and State Maintenance	Practical 3: Database Connection and User Authentication	

5 13/7/26 - 19/7/26	Interacting with the Server's Filesystem - File Handling Operations in PHP - Method for File Validation - File Upload Security Measures - Network Protocols in Data Transfer - Email Functionality in PHP	Practical 4: CRUD Functionality Implementation	
6 20/7/26 - 26/7/26	PHP Error Handling - PHP Exception Handling - Common PHP Error Types - PHP Error Reporting and Configuration - PHP Debugging Techniques - Custom Error Handling Techniques - Prevention and Strategies to Handle Errors	Practical 5: Cookies and Advanced Session Management	Quiz - Tentative Schedule (22/7/2026 (Wed))
7 27/7/26 - 2/8/26	Separating Content from Presentation - Separation of Concerns (SoC) Principle - Separating HTML Markup from PHP Logic - Model-View-Controller (MVC) architecture in PHP - Maintain Codebase through Content-Presentation Separation	Practical 6: File Upload and Processing	
8 3/8/26 - 9/8/26	Mid Term Test: 6/8/2026 (Thu)	Practical 7: PHP Error Handling	
9 10/8/26 - 16/8/26	AJAX - Asynchronous JavaScript and XML - AJAX Principles in Asynchronous Data Exchange - XMLHttpRequest Object and Functionalities - AJAX Integration in PHP - Server-Side Response Handling in AJAX - Features using AJAX for Real-Time Data Retrieval	Practical 8: Content-Presentation Separation	**14-15/8 (Fri-Sat) – UTAR Convocation
10 17/8/26 - 23/8/26	Securing Web Application - Introduction to Input Validation and Security - Form Validation Techniques - Common Web Security Threats - Security Measures to Prevent Common Vulnerabilities	Practical 9: AJAX Fundamentals	

	Fixing Authentication and Vulnerabilities		
11 24/8/26 - 30/8/26	Web Server Optimization - Introduction to Web Server Optimization - Quality Metrics and Optimization Strategies - PHP and Server Performance Optimization - Optimization Tools and Techniques - Analyzing and Resolving Performance Issues	Practical 10: Web Application Security	**25/8 (Tue) – Prophet's Birthday Assignment Submission: 24/8/2026 (Mon) by 5.00 PM
12 31/8/26 - 6/9/26	Hosting Environment and Advancements in Server-Side Technologies - Types of Hosting Environments - Web Application Hosting Strategies - Server-Side Technology Evolution - PHP Frameworks - MVC architecture in PHP Framework		**31/8 (Mon) – National Day Practical Test - Tentative Schedule (2/9/2026 (Wed))
13 7/9/26 - 13/9/26	Revision		
14 14/9/26 - 20/9/26	Revision		**16/9 (Wed) – Malaysia Day